Sustainability Concept & Strategy

Water, carbon, energy and shade — including where the scheme runs a deficit

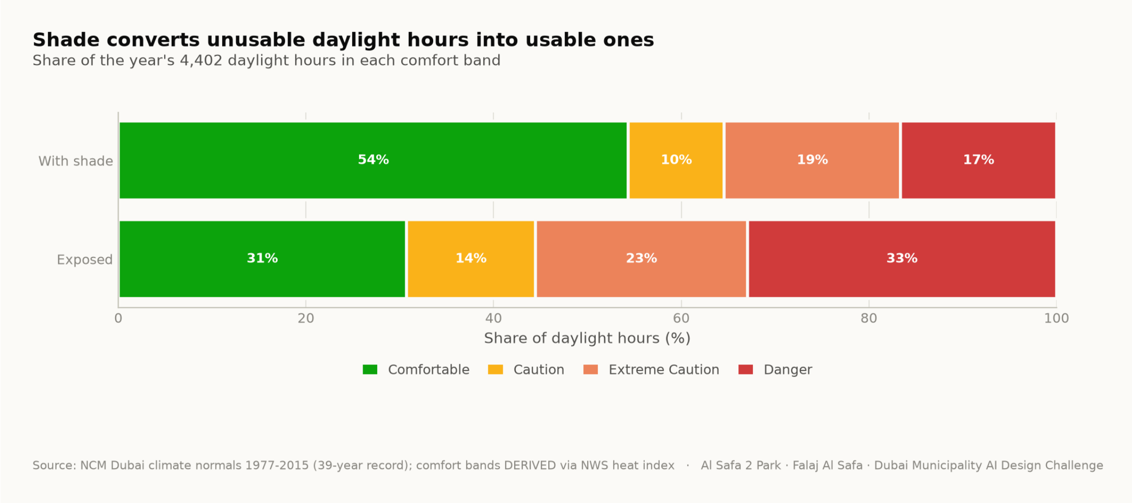
A sustainability chapter that reports only its successes is not evidence, it is marketing. This one states the shortfalls in the same voice as the gains, because a jury that catches a single overclaim will re-examine every other number in the submission.

Upload slot 08 — Sustainability Concept & Strategy
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 01 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Thermal performance — the primary outcome

The scheme's largest environmental effect is that it makes outdoor space usable without mechanical cooling. Comfortable daylight hours rise from 44.5% to 64.6%; the mean heat-index reduction under the canopy is 7.13 °C; and peak heat index falls from 56.8 °C to 48.7 °C.



Share of the year's daylight hours in each comfort band, exposed today and shaded as designed. Analysis output — computed from project data.

2. Water

Al Falaj is a 0.9 m recirculating channel — about 105 m² of water surface in total. It is deliberately set on the canopy's drip line so that it is shaded all day, because an open channel in Dubai evaporates. **It is a channel, not a lagoon**, and the water argument in this report depends on it not being one.

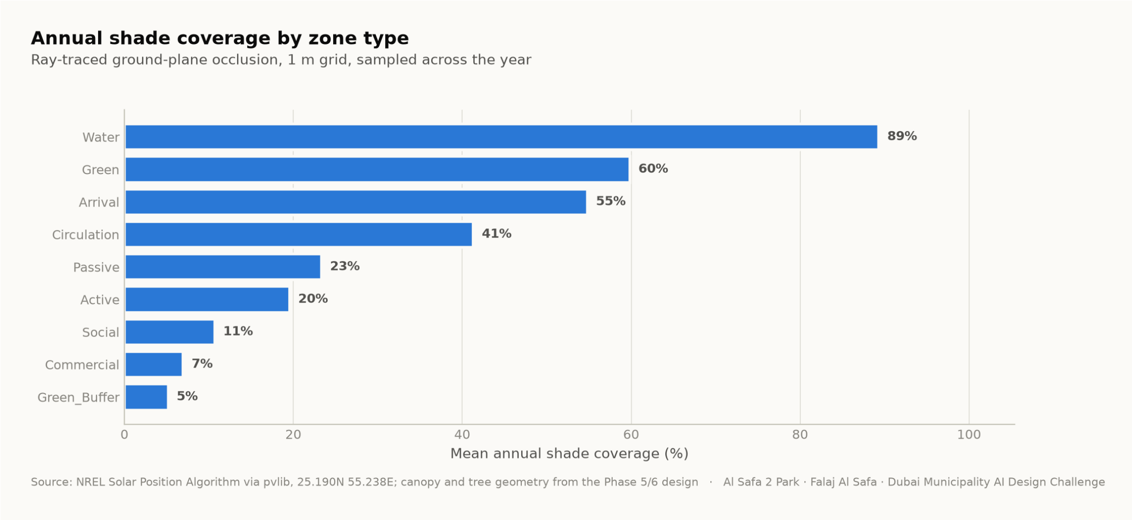
- Irrigation is subsurface drip fed from treated sewage effluent, not potable supply.
- Planting is five desert species selected for drought tolerance; Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.
- The falaj recirculates and is shaded for its whole length.

3. Energy — stated as the deficit it is

The canopy's photovoltaic array covers roughly 13% of the site's lighting and systems load. That is a deficit, not a surplus. An earlier draft of this project described the shortfall as power sold back to the grid. It is corrected here. The park is a net consumer of electricity. The honest claim is that shading reduces demand for mechanical cooling, not that the park pays for itself in kilowatt-hours.

4. Carbon, biodiversity and materials

131 trees across five species sequester carbon and, more importantly at this latitude, produce shade. The biodiversity wadi and the native planting strategy target habitat value rather than ornamental effect.



Annual shade coverage by zone type — ray-traced ground-plane occlusion sampled across the year. Analysis output — computed from project data.

5. What is conservative, and what is assumed

- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively — the real shaded condition is slightly better than reported, never worse.
- **The 6 °C shade relief is a literature value**, not a site measurement.
- **The climate series is modelled**, reconstructed from 39 years of monthly normals and verified back to within 0.39 °C. Conclusions are about a typical year, never about extremes.
- **Site-wide mean shade is 34.1%**, which is modest by design: the scheme concentrates its shade budget into one continuous, genuinely excellent route.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.